

Team name:	RescueBot IITA
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Name of the	Total Cost	Total Cost (\$)
Local Currency	Local Currency	U.S.A. Dollars
Argentine Peso	1,031,132.16	\$543.29
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Excha

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)	Status	Kit / Custom-built
1	Main PCB fabrication batch	Team-designed 2-layer PCB, 169.91 mm x 122.91 mm, batch of 5 boards	RescueBot IITA (team) / JLCPCB	JLCPCB Gerber quotation	1	22,889.88	\$12.06	22,889.88	\$12.06	New	Custom-built
2	Chassis & rescue mechanism (custom)	3D-printed PLA chassis, claw, mounts, storage and deposit guides	RescueBot IITA (team)	Self-made / local PLA estimate	1	35,000.00	18.44	35,000.00	\$18.44	New	Custom-built
3	Main computer	Raspberry Pi 4 Model B 8GB	Raspberry Pi Ltd	MercadoLibre	1	329,005.40	173.34	329,005.40	\$173.34	New	Commercial part
4	Microcontroller	Teensy 4.1	PIRC / SparkFun	SparkFun	1	59,787.00	31.50	59,787.00	\$31.50	New	Commercial part
5	Camera	USB wide-angle camera 140 deg (HBV-1716WA / ASIN B0DR7XXJL1)	Yosoo / HBVCAM	Tiendamia / Amazon mirror	1	47,670.00	25.12	47,670.00	\$25.12	New	Commercial part
6	IMU	BNO055 9-DOF Absolute Orientation IMU Breakout	Bosch / Adafruit	Adafruit	1	66,335.10	34.95	66,335.10	\$34.95	New	Commercial part
7	ToF distance sensor	VL53LOX module	Generic	MercadoLibre	2	10,989.00	5.79	21,978.00	\$11.58	New	Commercial part
8	Ultrasonic sensor	HC-SR04	Generic	MercadoLibre	3	2,940.00	1.55	8,820.00	\$4.65	New	Commercial part
9	Color/proximity sensor	APDS-9960 / GY-9960 RGB, gesture and proximity module	Broadcom / Avago / module vendor	MercadoLibre	1	8,090.00	4.26	8,090.00	\$4.26	New	Commercial part
10	Drive motor + encoder	DFRobot FIT0441 12V 159RPM brushless motor with encoder	DFRobot	DFRobot	4	37,770.20	19.90	151,080.80	\$79.60	New	Commercial part
11	Servo	DFRobot SER0056 2kg 300 deg clutch servo	DFRobot	DFRobot	5	11,388.00	6.00	56,940.00	\$30.00	New	Commercial part
12	Omniwheel	58mm aluminum omniwheel, 12kg load rating	Generic / Fulai-style	eBay equivalent	2	35,834.24	18.88	71,668.48	\$37.76	New	Commercial part
13	Fixed wheel	Pololu 1420 60x8mm wheel - unit cost split from pair price	Pololu	Pololu	2	6,405.75	3.38	12,811.50	\$6.75	New	Commercial part
14	Battery	CNHL LiPo 11.1V 3S 2200mAh 30/60C with XT60	CNHL	MercadoLibre	1	66,119.00	34.84	66,119.00	\$34.84	New	Commercial part
15	Buck regulator (compute rail)	XL4016 8A adjustable step-down, set to 5.0V	Generic (XL4016 module)	MercadoLibre equivalent	1	8,249.00	4.35	8,249.00	\$4.35	New	Commercial part
16	Buck regulator (servo rail)	MP1584 3A adjustable step-down, set to 6.1V	Generic (MP1584 module)	MercadoLibre	2	3,799.50	2.00	7,599.00	\$4.00	New	Commercial part
17	Power connector	XT60 male/female connector pair	Generic / Amass-style	MercadoLibre	1	3,545.00	1.87	3,545.00	\$1.87	New	Commercial part
18	Limit switches	FCL/FCR micro limit switches for deposit alignment	Generic / Candy-Ho	MercadoLibre	2	4,493.00	2.37	8,986.00	\$4.74	New	Commercial part
19	Main switch	12V ON/OFF power/start switch	Generic	MercadoLibre	1	2,610.00	1.38	2,610.00	\$1.38	New	Commercial part
20	Indicators	Buzzer + status LEDs	Generic	MercadoLibre packs	1	13,449.00	7.09	13,449.00	\$7.09	New	Commercial part
21	Relay module	5V 1-channel relay module	Generic / WEB3ARG	MercadoLibre	1	3,499.00	1.84	3,499.00	\$1.84	New	Commercial part
22	Wiring & fasteners	Cables, connectors, M3 screws, nuts, standoffs and heat-shrink	Various / local store	Local store estimate	1	25,000.00	13.17	25,000.00	\$13.17	New	Commercial materials / custom assembly

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#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	drivebase (custom)	Library	Custom motor/encoder drivebase control for the Teensy	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
2	claw (custom)	Library	Custom five-servo rescue mechanism control: pickup, sort and deposit	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
3	PID (custom/local)	Library	PID control for motor and movement regulation	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
4	Adafruit BNO055	Library	Driver for the BNO055 9-DOF IMU	github.com/adafruit/Adafruit_BNO055	Adafruit	FREE	FREE
5	Adafruit APDS9960	Library	Driver for the APDS-9960 color/proximity sensor	github.com/adafruit/Adafruit_APDS9960	Adafruit	FREE	FREE
6	Adafruit Unified Sensor / BusIO	Library	Sensor abstraction and I2C device layer	github.com/adafruit	Adafruit	FREE	FREE
7	VL53L0X (Pololu)	Library	Driver for the VL53L0X time-of-flight sensor	github.com/pololu/vl53l0x-arduino	Pololu	FREE	FREE
8	NewPing	Library	HC-SR04 ultrasonic helper library included locally for tests/examples	bitbucket.org/teckel12/arduino-new-ping	Tim Eckel	FREE	FREE
9	Servo (Arduino)	Library	Servo PWM control for rescue mechanism	arduino.cc	Arduino	FREE	FREE
10	elapsedMillis	Library	Non-blocking timing helper	github.com/pfereick/elapsedMillis	Paul Stoffregen / community	FREE	FREE
11	OpenCV contrib Python	Library	Image processing for line, marker and rescue-zone vision	pypi.org/project/opencv-contrib-python	OpenCV.org	FREE	FREE
12	NumPy	Library	Array and math operations	numpy.org	NumPy developers	FREE	FREE
13	pySerial	Library	UART communication with the Teensy	pyserial.readthedocs.io	pySerial developers	FREE	FREE
14	TensorFlow Lite Runtime	Library	On-device victim/zone inference	tensorflow.org/lite	Google	FREE	FREE
15	Ultralytics YOLOv8	Library	Rescue-detector training and export	ultralytics.com	Ultralytics	FREE	FREE
16	AGCWD + anti-flash preprocessing (custom)	Algorithm	Adaptive illumination correction for the 2026 LED-wall rule	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
17	CentroidTracker (custom)	Algorithm	Target stability/tracking between AI inferences	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
18	camthreader (custom)	Library	Threaded camera capture with latest-frame buffer	RescueBot IITA repo	RescueBot IITA (team)	FREE	FREE
19	Rescue victim/zone dataset + exported model (custom)	Dataset / AI model	Roboflow dataset and YOLOv8/TFLite/ONNX model trained by the team	Roboflow dataset - roboliga-2025-kh2ly	RescueBot IITA (team)	FREE	FREE
20	PlatformIO	Software's Tool	Teensy firmware build system	platformio.org	PlatformIO	FREE	FREE
21	Visual Studio Code	Software's Tool	Code editor	code.visualstudio.com	Microsoft	FREE	FREE

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22	Fusion 360	Software's Tool	3D CAD design of chassis and rescue mechanism	autodesk.com	Autodesk	FREE/EDUCATION	FREE/EDUCATION
23	EasyEDA	Software's Tool	PCB schematic and layout design	easyeda.com	EasyEDA / LCSC	FREE	FREE
24	Roboflow	Software's Tool	Dataset annotation and management	roboflow.com	Roboflow	FREE	FREE
25	Kaggle	Software's Tool	Model training and validation with cloud notebooks/GPU	kaggle.com	Kaggle / Google	FREE	FREE
26	Git / GitHub	Software's Tool	Version control and repository hosting	github.com	GitHub / Software Freedom Conservancy	FREE	FREE